

The response of the macroinvertebrate community to low-flow variability and supra-seasonal drought within a groundwater dominated stream

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With 5 figures and 3 tables

Abstract: Information regarding the instream response to long supra-seasonal droughts, extending over more than one season, in largely perennial riverine systems is fragmentary compared to knowledge regarding floods. The macroinvertebrate community of a groundwater dominated river was investigated over an eight-year period which included two supra-seasonal drought events (1992 and 1996–1997). The community responded directly to changes in the discharge regime. Significant differences in the community abundance, the abundance of the amphipod *Gammarus pulex*, diversity indices (Shannon Wiener and Simpson diversity) and the Berger-Parker dominance index were recorded between drought, flow recovery and other (non-drought) years. Drought conditions resulted in extremely low community abundance, *Gammarus pulex* abundance and the Berger-Parker dominance index, and extremely high values of the Shannon Wiener and Simpson diversity indices. Recovery of the community extended over a two-year period and reflected recovery of flow associated with recharge of the groundwater aquifer. The results underline the continuing need for medium to long-term data series to investigate the response of instream communities to flow variability so that extreme events can be placed within the context of natural variability.

Key words: Discharge variability, community change, groundwater, hydroecology, drought, macroinvertebrates.

Introduction

The importance of discharge (flow) variability within lotic ecosystems has increasingly been recognised (LAKE 2000, TOWNSEND et al. 1997), particularly

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the role of hydrological extremes (CLAUSEN & BIGGS 1997, RICHTER *et al.* 1997). Floods and droughts are both natural phenomena which occur frequently and predictably (e.g. annual floods or seasonal droughts) in many riverine ecosystems (CARUSO 2002, CLAUSEN & BIGGS 1997, POFF 1996, POFF 2002); although high magnitude events with relatively long return periods are less predictable in terms of their timing and duration (LAKE 2003). Research on the effects of extreme flows on instream communities has to date largely centred on the impact of high flow/discharge events (floods) (RESH *et al.* 1988, GILLER 1996, POFF 2002). In marked contrast, scientific research on the effects of extreme low flows, usually associated with drought, has lagged behind that on floods (HUMPHRIES & BALDWIN 2003). Most studies have concentrated on naturally ephemeral/intermittent streams that experience regular seasonal drying and rewetting, to which the organisms frequently display life-cycle adaptations (BOULTON & LAKE 1992, FEMINELLA 1996, WRIGHT *et al.* 1984).

Over 150 different definitions of drought have been identified in the scientific literature, reflecting the use of a variety of meteorological, hydrological and ecological criteria (AGNEW 2002, WILHITE 2000). The instream ecological response associated with these events varies greatly depending on the frequency, timing and duration of individual events, geographical coverage and permanence of flow at sites (LAKE 2000, WOOD & PETTS 1999). Drought may have markedly different impacts on instream communities across relatively small spatial scales reflecting natural variations in catchment characteristics such as stream order and groundwater storage. These differences may help to explain some of the conflicting results reported in the literature for many upland (e.g. HYNES 1958, MORRISON 1990, ERMAN & ERMAN 1995) and lowland streams and rivers (e.g. EXTENCE 1981, WRIGHT & BERRIE 1987, WOOD & PETTS 1999).

The response of instream communities to extreme discharge events reflects the life history characteristics of individual members of the community (e.g. COWX *et al.* 1984, ERMAN & ERMAN 1995, WRIGHT *et al.* 1981). Lotic ecosystems have a relatively low resistance to flow disturbances but their communities recover quickly due to their high resilience (MILNER 1994, LAKE 2000). However, recovery times may also reflect the magnitude (severity) and frequency (return period) of the event and natural variations in catchment characteristics (LAKE 2003).

The response of macroinvertebrate communities to low magnitude droughts, that extend over a single season, have been characterised by marked reductions in community abundance and diversity. Recovery following the drought is usually rapid over the following months in most natural impervious catchments with variable discharge regimes (CANTON *et al.* 1984, COWX *et al.* 1984, MORRISON 1990), although the impact may be modified in anthropogenically impacted (urbanised and regulated) systems (HUMPHRIES & BALDWIN 2003).

In contrast, the impact of seasonal droughts has been difficult to detect within most groundwater dominated streams and rivers due to the buffering capacity of the aquifer (BICKERTON 1995, SEAR et al. 1999). However, droughts that extend over more than one season, termed “seasonal or supra-seasonal” droughts by LAKE (2003), particularly over the winter months when groundwater aquifers are recharged, have the potential to lead to significant changes in instream community structure (ARMITAGE & PETTS 1992, WRIGHT & SYMES 1999, WRIGHT et al. 2002).

The current investigation examines the effect of natural low flow variability over an eight-year period on the instream macroinvertebrate community in the upper reaches of a small perennial chalk stream in south-east England. Two spatial scales are examined: i) nine sites encompassing all habitat types (e.g. pool, riffle and run) on the river; and ii) a sub-set of four riffle sites – these habitats represent morphological high points on the riverbed which are known to support distinctive communities and are sensitive to changes in discharge/flow-depth (BOULTON 2003, BRUSOCK & BROWN 1991). The study period included two high magnitude supra-seasonal droughts (1989–1992 and 1996–1997) followed by periods of natural flow recovery. We hypothesise that (i) the diversity and abundance of the macroinvertebrate community will reflect the flow conditions experienced over the 8-year period; and (ii) recovery following supra-seasonal droughts will be extended over a greater period of time (in excess of 12-months) than those reported for short duration events (e.g. EXTENCE 1981, MORRISON 1990).

Study site

The Little Stour River is 11.5 km long, with a catchment area of approximately 213.3 km², rising 4 km east of the city of Canterbury, Kent, England. Along with the ephemeral Nailbourne River and the perennial Wingham River, the Little Stour forms the only tributaries of the Great Stour River (See WOOD & PETTS 1999 for detailed site characteristics and map). The underlying geology of the catchment is chalk although this is overlain by substantial gravel deposits in the mid and lower-reaches of the river. The upper river is typical of other chalk streams and is characterised by fast flow velocities (typically $>0.40 \text{ m s}^{-1}$), substantial seasonal macrophyte growths, and diverse instream habitat patches (BERRIE 1992). The channel is narrow (5–10 m wide) and has a mixed substratum of gravel, sand, and silt interspersed with macrophyte stands. The macroinvertebrate community of the headwaters is naturally diverse and dominated by the amphipod *Gammarus pulex*, net spinning Trichoptera from the family Hydropsychidae, and the cased caddisfly *Agapetus fuscipes* (Glossosomatidae) (WOOD & ARMITAGE 1999). Below the confluence with the Wingham River the river has been embanked and the bed and banks are regularly dredged to prevent flooding of adjacent agricultural land (WOOD et al. 2000). The lower reaches of the river are characterised

by taxa indicative of lentic and slow-flowing environments, particularly Gastropoda and Coleoptera, with abundant macrophyte growths (WOOD & PETTS 1999). The sites from the lower reaches have been excluded from the current investigation due to the highly regulated nature of the instream and riparian habitats. For further details of specific site locations and physical characteristics, including relative permanence, see WOOD & PETTS (1999) and WOOD *et al.* (2000).

Flow history

The discharge regime of the Little Stour is characterised by flow stability and an absence of short-term peaks associated with storm events. The hydrograph displays a clear annual rise and fall of discharge typical of chalk and many other groundwater dominated streams. Peak discharge usually occurs between December and February and base-flow usually occurs between July and September each year (Fig. 1). During the study period the river experienced two periods of severe low flow due to supra-seasonal droughts (1989–1992 and 1996–1997), leading to the desiccation of two historically perennial reaches. Due to the extended nature of these droughts no isolated pools persisted throughout either period. Re-wetting of desiccated reaches occurred within two months of the end of the drought in late 1992 and 1997, respectively. However, due to low groundwater levels within the catchment, recovery of discharge following both droughts was extended over the following 12–18 months. For the purpose of analysis, the annual flow series has been classified into three groups: drought (1992, 1996 and 1997), flow recovery (the year following drought) (1993 and 1998) and other (non-drought) years (1994, 1995 and 1999) (Fig. 1).

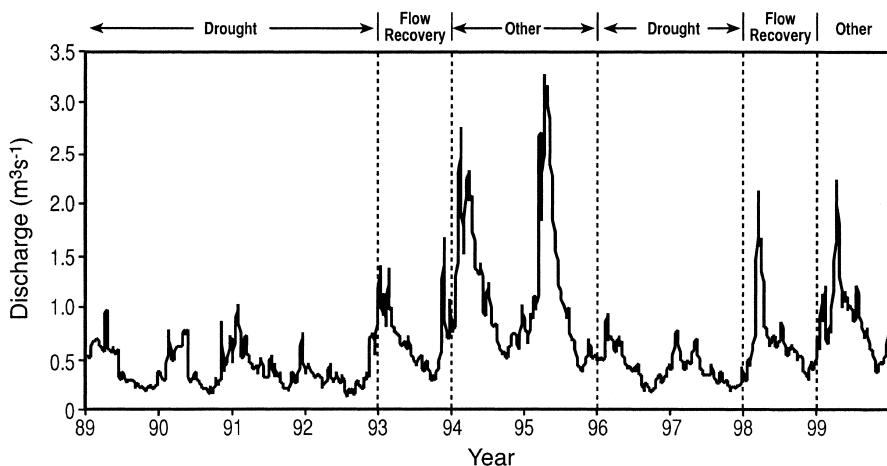


Fig. 1. Discharge hydrograph for the Little Stour River (Jan. 1989–Dec. 1999) indicating the drought, flow recovery and other (non-drought) periods. Tick marks on X-axis indicate January 1st of each year.

Methods

Macroinvertebrate samples were collected from 9 historically perennial sites during the first week of September between 1992 and 1999, coinciding with the typical period of lowest discharge within the catchment. Samples were collected using two-minute semi-quantitative kick samples with a standard pond net (250 μm mesh net) covering all of the available habitats within a 20 m reach at a site. The use of the same personnel throughout the study period ensured that comparisons could be made between ecological data for each site and between years throughout the series (FURSE et al. 1981). Samples were preserved in the field with 70 % industrial methylated spirits prior to processing in the laboratory. Consistent sampling dates were used to facilitate comparison between similar stages in faunal life histories and the hydrological year. A total of 66 macroinvertebrate samples was collected; 6 could not be retrieved due to the desiccation of 3 sampling sites during drought conditions in 1992 and 1997, respectively. Taxa were identified to species level where possible, although a number of taxa, including Baetidae, Chironomidae and Oligochaeta, were left at a higher taxonomic level due to time constraints; while others such as Corixidae, Notonectidae and Pyralidae were confined to a small number of samples and occurred in low abundances. These latter taxa probably invaded the upper river from the regulated and ponded lower reaches during periods of low flow and drought.

Invertebrate community data were analysed using correspondence analysis (CA) within the programme CANOCO (TER BRAAK & ŠMILAUER 1998) to graphically assess differences between samples collected during drought years, the flow recovery and other (non-drought) years. All data were transformed ($\log_e + 1$) prior to correspondence analysis to reduce the clustering of common and abundant taxa at the centre of the ordination plot, and therefore allow the examination of structural differences in community composition.

To examine the influence of low flow variability on the macroinvertebrate community the number of taxa, log-transformed relative community abundance and log-transformed relative abundance of *Gammarus pulex* (the most abundant and widely distributed taxon in the river) were calculated. In addition, the Shannon-Wiener diversity index, Simpson diversity and the Berger-Parker dominance index were calculated, using the α Species Diversity and Richness software (Pisces Conservation 1998), to examine changes in the structure of the macroinvertebrate community. These variables were used as dependent variables within a one way analysis of variance (ANOVA). Post-hoc Tukey's test was used to identify any significant differences between years of drought (1992, 1996 and 1997), the flow recovery years (1993 and 1998), and other years (1994, 1995 and 1999). This ensured correction of experimentwise errors and reduced the potential for spurious significant associations.

To examine the influence of discharge variability on an individual species at different spatial scales, the log-transformed standardised relative abundance of the amphipod, *Gammarus pulex*, from all nine sites and for the four riffles on the river was analysed in detail. Twenty-three discharge variables, up to 12 months prior to sample collection, were derived (Table 1). Correlation coefficients were computed to identify any seasonality and periods of increased association between discharge and the abundance

Table 1. Discharge variables used in correlation and regression analysis of the Little Stour River.

Discharge variables	Definition	Number
D- <i>n</i>	Mean daily discharge measured 1–7 (<i>n</i>) days prior to sampling	7
M- <i>n</i>	Mean daily discharge for 1–12 (<i>n</i>) months prior to sampling	12
Y-3	Mean daily discharge over the 3–months prior to sampling	1
Y-6	Mean daily discharge over the 6–months prior to sampling	1
Y-12	Mean daily discharge over the 12–months prior to sampling	1
DWF	Dry Weather Flow index (average of the seven consecutive days of lowest discharge in the 12-months prior to sampling)	1

of *G. pulex* within the series. These variables included discharge characteristics in the 7-days prior to sample collection (D-*n*), average daily discharge characteristics for each of the 12-months prior to sample collection (M-*n*), and averaged daily flows over 3-month (Y-3), 6-month (Y-6) and 12-month (Y-12) lagged periods. Scatter plots were produced and inspected to examine the nature of relationships (to ensure a linear relationship) prior to subsequent analysis. These variables were then offered as candidate predictors in a stepwise multiple linear regression for (i) all sites; and (ii) the sub-set of four riffle sites.

Ecological variables were standardised for computation of correlation coefficients and regression analysis, and production of box-plots, by calculating z-scores for each individual sample site within the series 1992–1999 (mean = 0 and standard deviation = 1). This method of standardisation, undertaken prior to analysis, re-scales data from individual sites to enable comparisons between the responses of multiple sites to the same external factor (discharge variability), and is widely used in other areas of aquatic sciences (e.g. HARRIS et al. 2000). This method does not alter the time series shape at individual sites or correlation coefficients with independent variables.

Results

A total of 87 taxa was recorded during the study period, ranging from 42 in 1992 to 60 in 1998, a 43 % increase from the period of lowest discharge. The most abundant species was the freshwater amphipod, *Gammarus pulex* (L.), which reached a density of 7469 m⁻² in 1994. Correspondence Analysis (CA) indicated that groups of sample sites and taxa could be identified visually based on the three discharge groups identified (Fig. 2), although overlap between the groups occurred, reflecting individual site characteristics. Samples from both periods of drought (1992, 1996 and 1997) formed a cluster predominately on the negative quadrants of axis 2. Samples collected during other (non-drought) years (1994, 1995 and 1999) were predominately located on the negative quadrants of axis 1 (Fig. 2 a), with the samples from the flow recovery

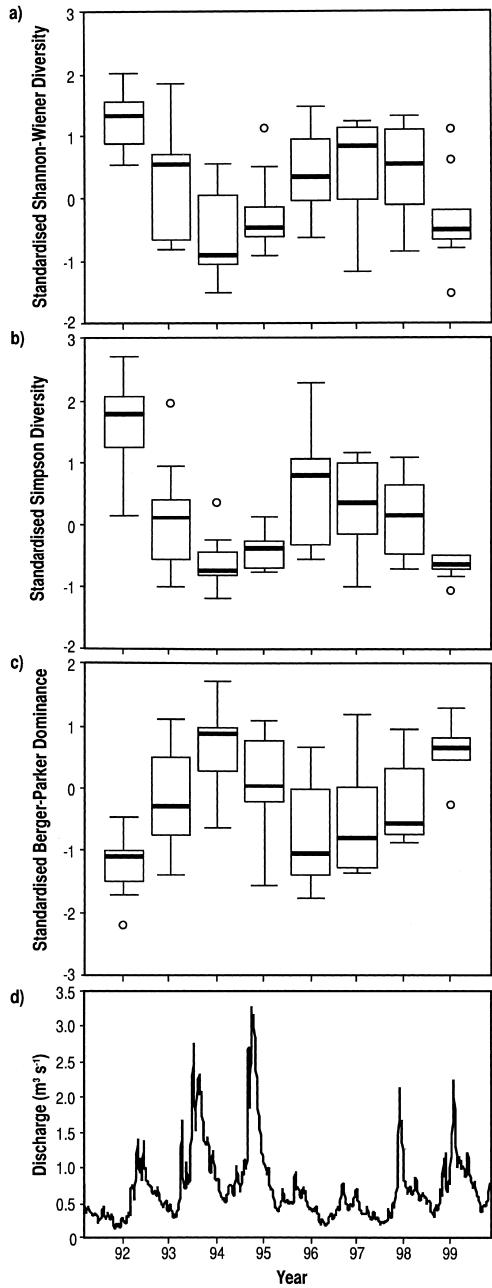


Fig. 3. Box-plots of standardised invertebrate community indices 1992–1999 for the Little Stour River (a) Shannon-Wiener diversity index, (b) Simpson diversity index, (c) Berger-Parker dominance index, and (d) discharge series. Solid line at centre of box indicates median value, box encapsulates interquartile range and outer lines span the range. O = outliers.

years (1993 and 1998) forming a widely dispersed group between the other two.

The distribution of the taxa on the taxon biplot reflected individual preferences for flow velocity conditions rather than the discharge categories identified. Taxa indicative of high-flow velocities were associated with the negative quadrants of axis 1 e.g. Trichoptera (*Hydropsyche siltalai*, *Rhyacophila dorsalis*, *Agapetus fuscipes* and *Drusus anulatus*), Simuliidae and Elmidae (*Elmis aenea*, *Riolus subviolaceus* and *Oulimnius* spp.) (Fig. 2 b). Taxa associated with low-flow velocities and in some instances lentic conditions, such as Coleoptera from the family Haliplidae (6 species), the dytiscid *Nebroporus depressus elegans* and Gastropoda (e.g. *Planorbis planorbis* and *Valvata piscinalis*) were almost exclusively located on the positive quadrants of axis 1.

Macroinvertebrate community structure and abundance standardised parameters appeared to respond directly to changes in the discharge regime (Fig. 3 and Fig. 4). The diversity indices (Shannon-Wiener and Simpson – Fig. 3 a and 3 b) had an inverse relationship with discharge (Fig. 3 d) whilst the Berger-Parker dominance index (Fig. 3 c), the number of taxa, log-community abundance and log-*Gammarus pulex* abundance (Fig. 4 a–c) had a positive relationship with discharge variability (Fig. 4 d). Drought conditions in 1992, 1996 or 1997 resulted in extremely low number of taxa, log community abundance, log-*G. pulex* abundance and the Berger-Parker dominance index and extremely high values of the Shannon-Wiener and Simpson diversity indices. However, with the exception of the standardised number of taxa, the highest discharge prior to sampling in 1995 did not result in extreme values of any indices.

One-way ANOVA indicated significant differences between the following faunal parameters for all three discharge groups (drought, flow recovery and other years) (Table 2 a): Shannon-Wiener diversity index, Simpson diversity, Berger Parker dominance index, log community abundance, log-*G. pulex* abundance (all $P < 0.005$). Post-hoc analysis (Tukey's tests) indicated significant differences were recorded between community parameters from drought and flow recovery years, with the exception of the Shannon-Wiener diversity index and the number of taxa, and for drought and other (non-drought) years, with the exception of the number of taxa (Table 2 b). In addition, there were significant differences recorded between the community indices for the flow recovery and other (non-drought) years, except the Simpson diversity index, Berger-Parker dominance index and the number of taxa (Table 2 b).

Pearson correlation coefficients for discharge and standardised log-*G. pulex* abundance data indicated that the strength of the association was greater for the four riffle sites than for all sites (Table 3). With the exception of the long-term flow variables for all sites (Y-3, Y-6 and Y-12), the discharge parameters had a significant relationship with the standardised log-*G. pulex* abun-

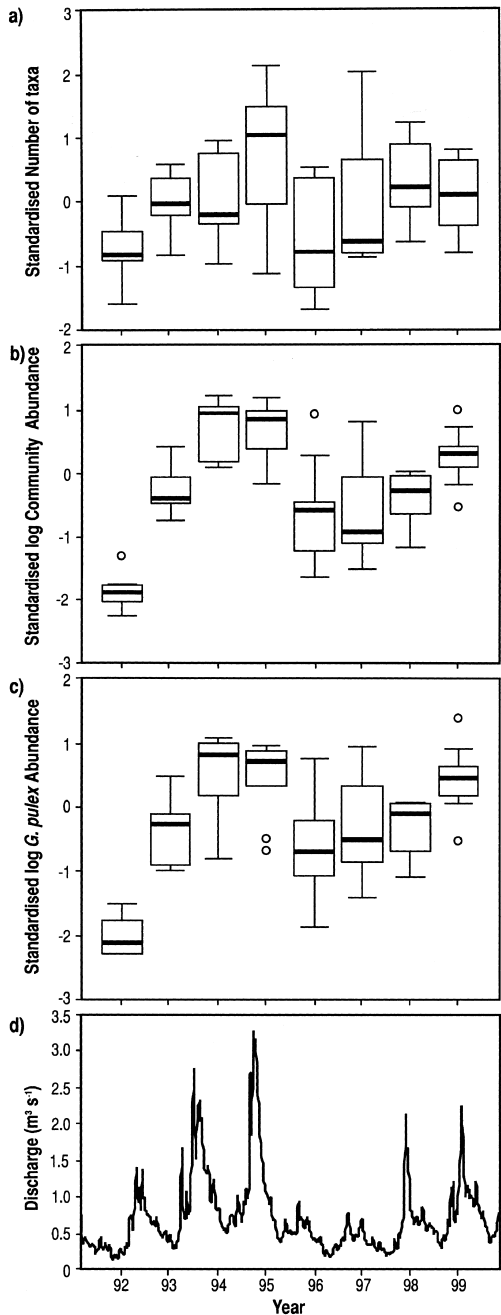


Fig. 4. Box-plots of standardised invertebrate community indices 1992–1999 for the Little Stour River (**a**) number of taxa, (**b**) log community abundance, (**c**) log-*G. pulex* abundance, and (**d**) discharge series. Solid line at centre of box indicates median value, box encapsulates interquartile range and outer lines span the range. O = outliers.

Table 2. One-way analysis of variance of macroinvertebrate community indices between (a) discharge categories; and (b) Post-hoc Tukey's test for drought and flow recovery years, drought and other years, and flow recovery and other years.

a)	df	Mean Square	F-ratio
Shannon-Wiener	2	2.68	12.10**
Simpson	2	52.25	13.57**
Berger-Parker	2	0.41	14.39**
Number of taxa	2	72.10	1.82
Log-abundance	2	17.86	24.61**
Log- <i>G. pulex</i> abundance	2	37.10	21.46**
b)	Mean difference between groups (i–ii)		
	(i) Drought and (ii) Recovery	(i) Drought and (ii) Other	(i) Recovery and (ii) Other
Shannon-Wiener	0.26	0.65**	0.39*
Simpson	1.95*	2.87**	0.92
Berger-Parker	–0.14*	–0.25**	–0.11
Number of taxa	–2.77	–3.23	–0.46
Log-abundance	–0.78*	–1.69**	–0.91*
Log- <i>G. pulex</i> abundance	–1.20*	–2.44**	–1.23*

* = $P < 0.05$; ** = $P < 0.005$.

dance, although they were all relatively weak when all of the sites were considered. A period of increased strength of Pearson correlation coefficients within the monthly averaged daily flow series 4–7 months (May–February and see Fig. 5 a and b) prior to sampling could clearly be identified for the four riffle sites (Table 3). This period of increased strength of the relationship directly corresponded with peak discharges recorded in the annual series for the catchment. This pattern was not as clear within the series for all nine sites and an extended period of significant association 1–8 months (January–August) was observed.

Correlation coefficients for discharge characteristics and standardised log-*G. pulex* abundance data for the month before sample collection (M-1) (Fig. 5 c) and immediately prior to sample collection (D-1) were weaker than those for seven days (D-7) prior to sample collection for all nine sites and the four riffle sites (Table 3). The relationship between the standardised log-*G. pulex* abundance and the Dry Weather Flow index (DWF) was stronger for the four riffle sites (Fig. 5 d) than for all sites. The development of stepwise multiple linear regression models identified M-4 ($R^2 = 0.28$, $P < 0.001$) for all sites and M-7 ($R^2 = 0.82$, $P < 0.001$) for riffle sites as the only predictors of the abundance of *G. pulex*. No other discharge parameters were incorporated into either model.

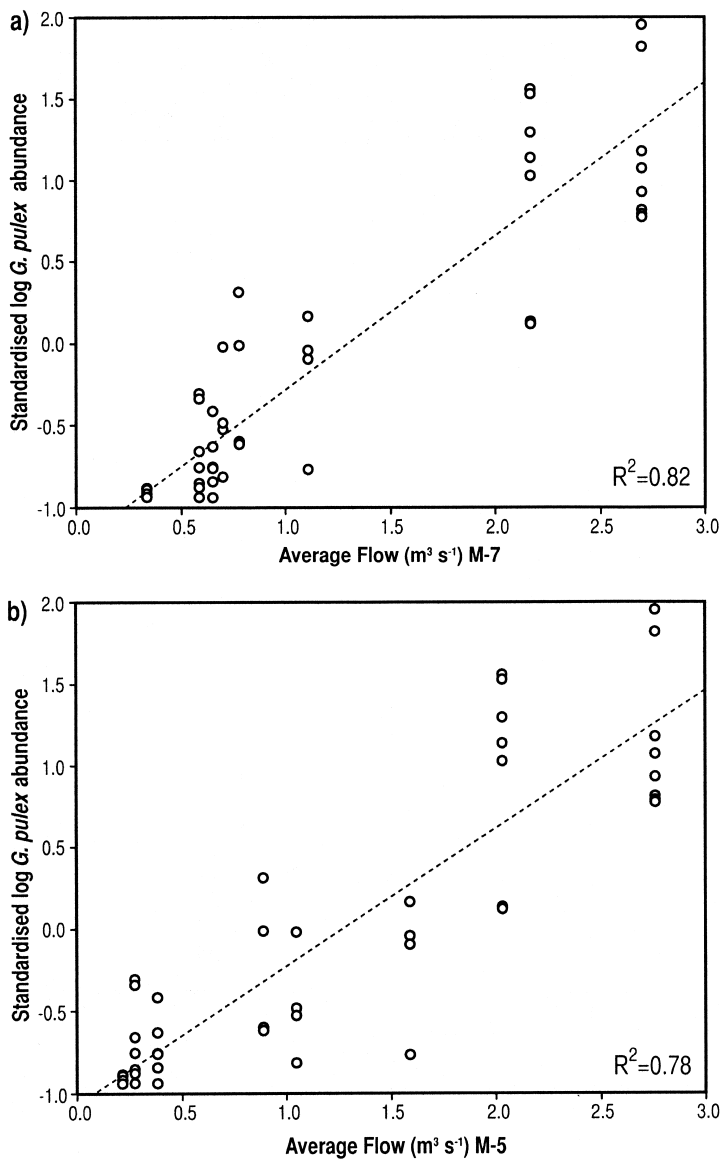


Fig.5. a and b.

Discussion

The 8-year study of the Little Stour River is unusual because the study period includes two periods of extended drought with data for the years between them. Few similar long-term studies have been undertaken, and where data

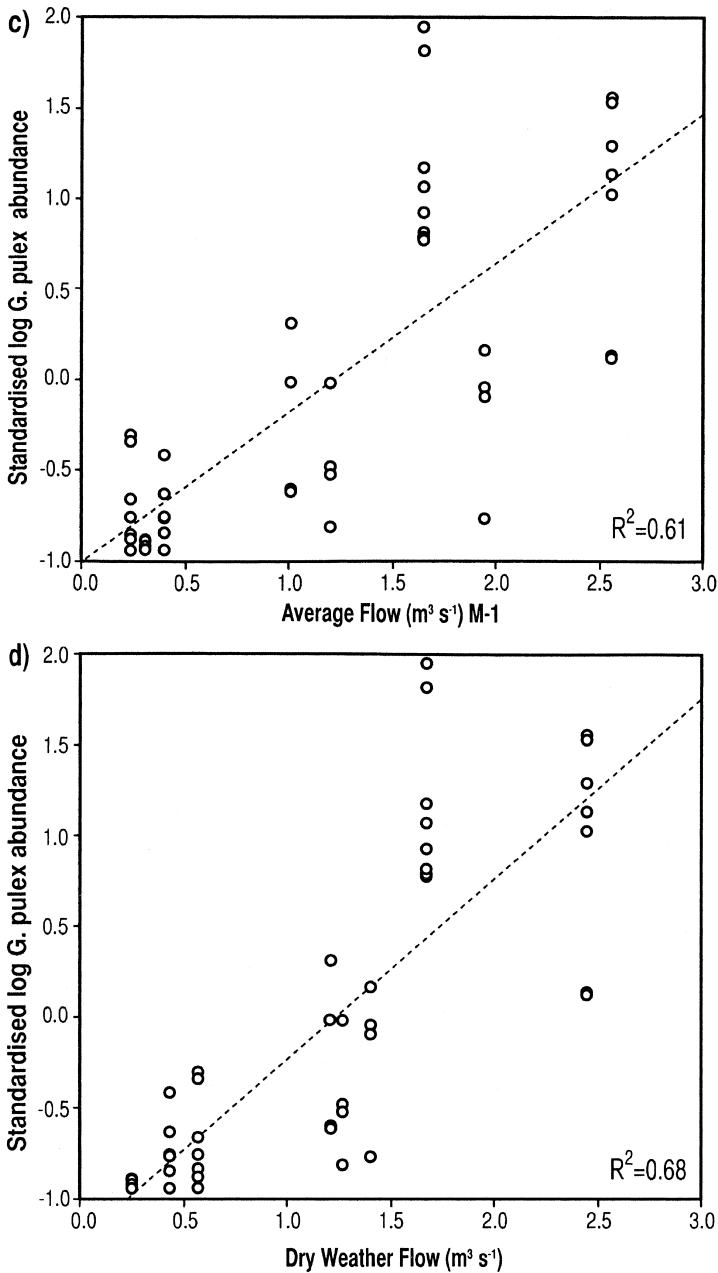


Fig. 5. Relationship between selected discharge variables and standardised log-*G. pulex* abundance from riffle sites on the Little Stour River: (a) mean daily discharge 7 months prior to sample collection – M-7, (b) mean daily discharge 5 months prior to sample collection – M-5; (c) mean daily discharge 1 month prior to sample collection – M-1, and (d) Dry Weather Flow index.

Table 3. Pearson correlation coefficients between the standardised log-*Gammarus pulex* abundance and discharge variables for all sites and for riffle sites on the Little Stour River (1992–1999).

Variable	All Sites (n = 66)	Riffle Sites (n = 30)
D-1	0.30*	0.41*
D-2	0.30*	0.38*
D-3	0.34*	0.46*
D-4	0.29*	0.46*
D-5	0.36*	0.51*
D-6	0.43**	0.58*
D-7	0.44**	0.66**
M-1	0.45**	0.78**
M-2	0.47**	0.82**
M-3	0.50**	0.83**
M-4	0.53**	0.89**
M-5	0.50**	0.88**
M-6	0.48**	0.89**
M-7	0.45**	0.91**
M-8	0.47**	0.84**
M-9	0.23	0.72**
M-10	0.44**	0.69**
M-11	0.32*	0.77**
M-12	0.34*	0.53*
Y-3	0.03	0.47*
Y-6	0.08	0.54*
Y-12	0.07	0.50*
DWF	0.42**	0.82**

* = $P < 0.05$; ** = $P < 0.005$.

See Table 1 for definition of variables.

does exist the study period frequently includes periods (up to ten-years) with little or no data, during which considerable natural and anthropogenic change may have occurred (e.g. BRADLEY & ORMEROD 2001, WRIGHT et al. 2003, WOODWARD et al. 2002).

Analysis of the macroinvertebrate community of the Little Stour with Correspondence Analysis (CA) identified three relatively broad groups, with considerable overlap, based on the three discharge categories identified (Fig. 2). Despite the marked hydrological variability, instream habitat diversity was maintained throughout the study period even during the severe drought conditions recorded during 1992 and 1997. As a result the distribution of the taxa on the ordination bi-plot appears to reflect general preferences for flow velocity (fast versus slow) rather than discharge volume. Biodiversity (number of taxa) for the upper river was reduced during periods of drought and was more pro-

nounced than for the downstream regulated sites (BOULTON 2003, WOOD & PETTS 1999).

A range in invertebrate community parameters (abundance, diversity and dominance) derived for the Little Stour respond directly to changes in the discharge regime of the river (Fig. 3 and Fig. 4). Significant differences were recorded between community parameters from drought, flow recovery and other (non-drought) years (Table 2). However, while the lowest discharge conditions (which coincided with the 1992, 1996–1997 drought periods) resulted in extreme values of the community indices recorded, the highest discharges recorded during the study period in 1995 resulted in extreme values for only one of the indices examined (number of taxa). This suggests that the sequence of flows, and the transition between drought, flow recovery and non-drought conditions, is probably as important as the total volume of flow in many groundwater dominated rivers (WOOD et al. 2001).

The amphipod *Gammarus pulex* is clearly a keystone species since it is a major determinant of the patterns in community structure observed in the Little Stour River (Fig. 3 and Fig. 4). Examination of correlation coefficients for *G. pulex* and the flow series identified periods prior to sample collection when the strength of the relationships were greater (Table 3). The timing of flow variation is central to the organisation of the instream community and faunal life cycles may be timed to either avoid or exploit periods of different magnitude (CLAUSEN & BIGGS 1997, RICHTER et al. 1997). In the Little Stour, the development of simple linear regression models for all sites and four riffle sites identified the averaged daily discharge 4 and 7 months (M-4 and M-7) prior to sample collection as the most significant variables respectively. This suggests that periods of high discharge, or their absence during periods of drought, are critical to the abundance of taxa such as *G. pulex*. These variables were selected in preference to other short and longer-term variables offered as candidate predictors and suggest significant redundancy in the hydrological data due to autocorrelation of variables (OLDEN & POFF 2003). In addition, the relatively weak relationships recorded when all of the sites were considered indicates the highly variable response of individual sites and the need to consider additional physical and biological factors in future investigations.

Research on the recovery of instream macroinvertebrate fauna following drought has reported relatively rapid recovery times ranging from approximately one month (LEDGER & HILDREW 2001, SAGAR 1983) to several months (COWX et al. 1984, MORRISON 1990) for impervious catchments and over a year in groundwater dominated streams (LADLE & BASS 1981, WRIGHT et al. 2002). The results from the Little Stour suggest that most of the community parameters examined for the drought, flow recovery and other (non-drought) years were significantly different (Table 2). This suggests that recovery of the macroinvertebrate community occurred over a two-year period because of the

extended nature of the recovery of discharge associated with the groundwater aquifer. This is almost certainly the result of the extended (supra-seasonal) nature of both droughts during the study period. Both droughts had their origin in low winter rainfall followed by warm dry summer periods (MARSH & TURTON 1996, MARSH 2001).

Comparison of the results recorded in this study with other recent research on severe low flows facilitates the identification of several distinct types of in-stream response to drought. These clearly need to be recognised and considered when reporting and comparing the results of individual studies. First, the response in naturally ephemeral/intermittent streams (BOULTON & LAKE 1992) is markedly different to largely perennial systems (BOULTON 2003). Second, there are marked differences in instream responses to predictable seasonal or annual events and longer supra-seasonal droughts (LAKE 2003). However, a third distinction, based on the contribution of flow derived from surface water or groundwater sources should be included (WOOD & PETTS 1999). The presence of an underlying groundwater aquifer provides a short to medium-term buffer against rainfall and river discharge variability and as a result there may be a lag-time from the onset of a meteorological drought (rainfall deficit) until the impact on the instream community is detected.

The three criteria above help to explain the marked difference in the results of this study and those of LEDGER & HILDREW (2001) for the surface water dominated Lone Oak Stream, parts of which became dry during the summer seasonal drought of 1995. Despite the geographical proximity of approximately 75 km between the two sites, the Little Stour was buffered from the effects of this high magnitude seasonal drought due to high rainfall and groundwater levels during the preceding winter of 1994 (MARSH & TURTON 1996). As a result the Little Stour experienced the highest summer discharges within the series during 1995 despite exceptionally low summer rainfall. It was only when the drought progressed to a supra-seasonal event in the summer of 1996 that the full impact of the drought on the macroinvertebrate community became apparent.

Conclusion

The results from the current investigation provide a valuable insight to the response of the macroinvertebrate community within a groundwater dominated river to flow variability over an eight-year period which included two supra-seasonal droughts. The macroinvertebrate community of the Little Stour River responded directly to changes in the annual flow regime. Severe low flows associated with drought resulted in low macroinvertebrate community abundance and Berger-Parker dominance index, and high Shannon-Wiener and

Simpson diversity indices. Significant differences were recorded between community parameters for drought years, flow recovery years and other (non-drought) years. These results suggest that recovery of discharge and the macroinvertebrate community following drought occurred over a two-year period; although the annual sampling regime does not enable the determination of the exact timing of recovery. This recovery period is longer than reported for other droughts in the UK (e.g. EXTENCE 1981, LEDGER & HILDREW 2001, WRIGHT et al. 2000) and elsewhere (BOULTON 2003); but reflects the supra-seasonal nature of both droughts within the study period and the significant contribution of groundwater to the discharge regime.

The number of datasets available to examine the ecological response of in-stream communities to low flows associated with drought is limited compared to those for floods. Much of the current knowledge regarding the ecological response to drought centres on the effects of largely predictable seasonal droughts rather than supra-seasonal events (LAKE 2003). Due to the extended nature of supra-seasonal events it is difficult to predict their occurrence and to date relatively few studies have been designed specifically to examine the in-stream response to drought (BOULTON 2003). To compound this problem the absence of appropriate pre-drought data in a number of studies has meant the assessment of the end point of recovery has not been possible. There is a continuing need for long-term data sets, such as the Little Stour, to be compiled and maintained to allow the assessment of the impact of these supra-seasonal events (WRIGHT et al. 2000) within the context of basic ecological understanding, water resource management and global climatic change.

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